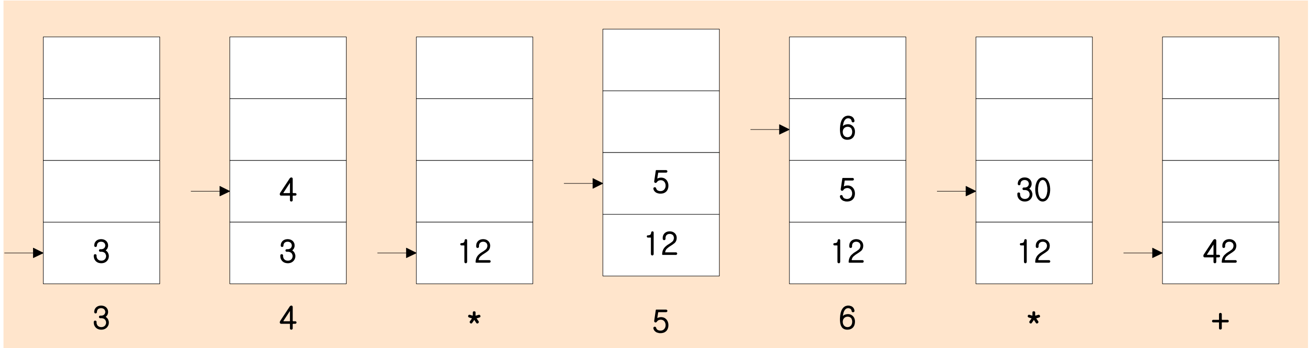
Stack Arithmetic

- RPN (*Reverse Polish Notation*)
  - The common mathematical method of writing arithmetic expressions imposes difficulties when evaluated by a computer
  - A stack organization is very effective for evaluating arithmetic expressions
  -) **A * B + C * D** → **AB * CD * +** : ***Fig. 8-5***
    - (3 * 4) + (5 * 6) → 34 * 56 * +



- 8-4 Instruction Formats
  - **Fields in Instruction Formats**

    1) **Operation Code Field** : specify the operation to be performed

    2) **Address Field** : designate a memory address or a processor register

    3) **Mode Field** : specify the operand or the effective address (*Addressing Mode*)

- 3 types of CPU organizations

X = Operand Address

  - 1) Single AC Org. : **ADD X** $AC \leftarrow AC + M[X]$
  - 2) General Register Org. : **ADD R1, R2, R3** $R1 \leftarrow R2 + R3$
  - 3) Stack Org. : **PUSH X** $TOS \leftarrow M[X]$ ADD

- Most of the modern systems use the Hybrid Approach.

- The influence of the number of addresses on computer instruction

  - **X = (A + B)*(C + D)**
    - 4 arithmetic operations **: ADD, SUB, MUL, DIV**
    - 1 transfer operation to and from ***memory*** and ***general register*** **: MOV**
    - 2 transfer operation to and from ***memory*** and ***AC register*** **: STORE, LOAD**
    - Operand memory addresses **: A, B, C, D**
    - Result memory address **: X**

  - 1) Three-Address Instruction

ADD	**R1, A, B**	$R1 \leftarrow M[A] + M[B]$
ADD	**R2, C, D**	$R2 \leftarrow M[C] + M[D]$
MUL	**X, R1, R2**	$M[X] \leftarrow R1 * R2$

- Each address fields specify either a processor register or a memory operand
  - **+** Short program
  - **-** Require too many bit to specify 3 address